

From LLMs to reasoning models

Agentic AI, part 1

Carlos Cotrini

September 2026

1

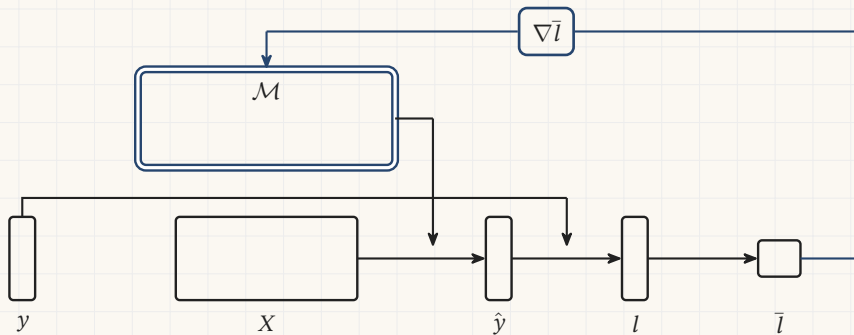
Section 1

How a language model is trained

The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

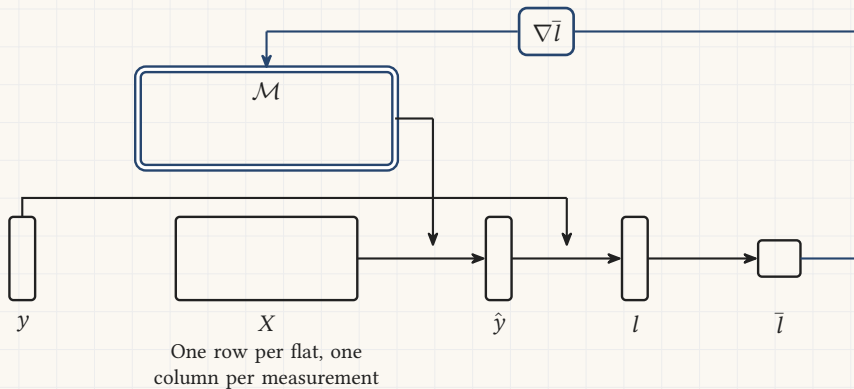
The five flats and the network are the ones from weekend 1.



The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

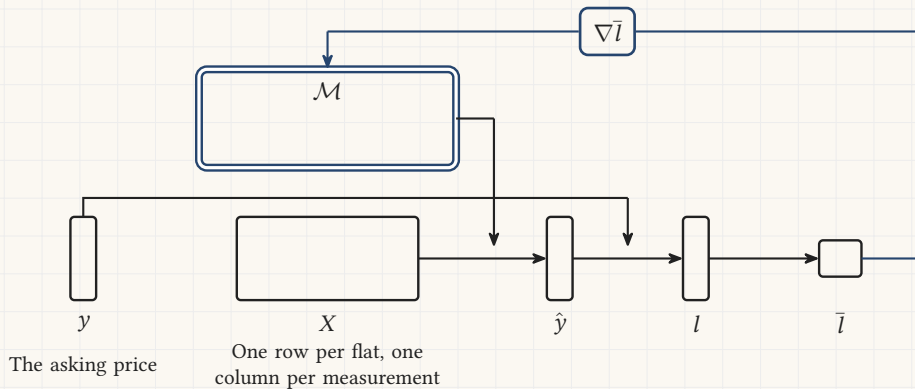
The five flats and the network are the ones from weekend 1.



The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

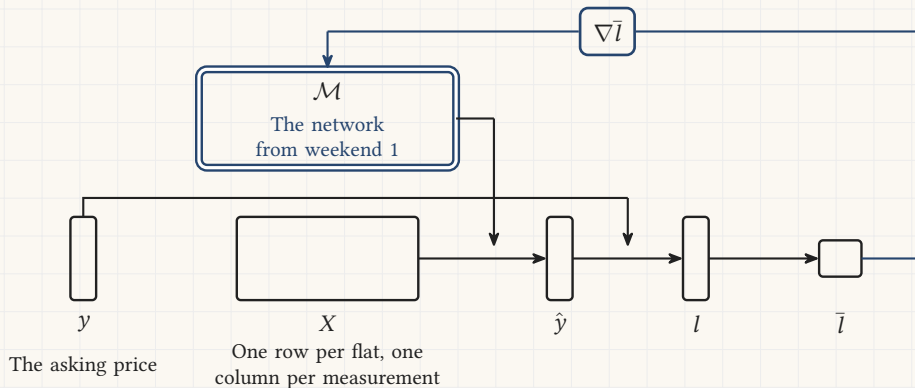
The five flats and the network are the ones from weekend 1.



The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

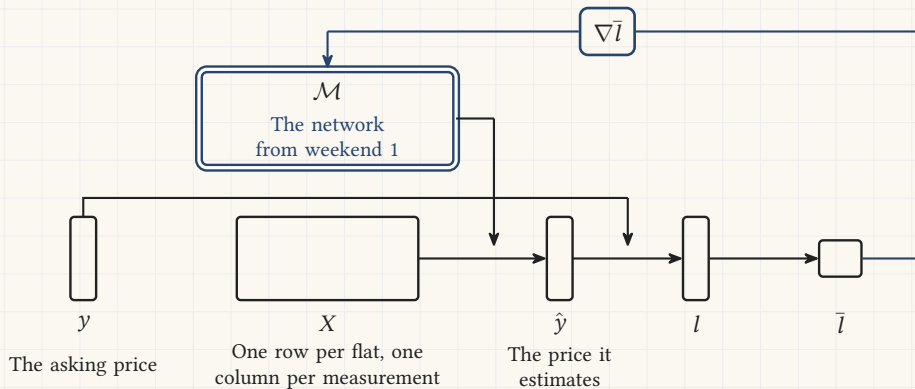
The five flats and the network are the ones from weekend 1.



The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

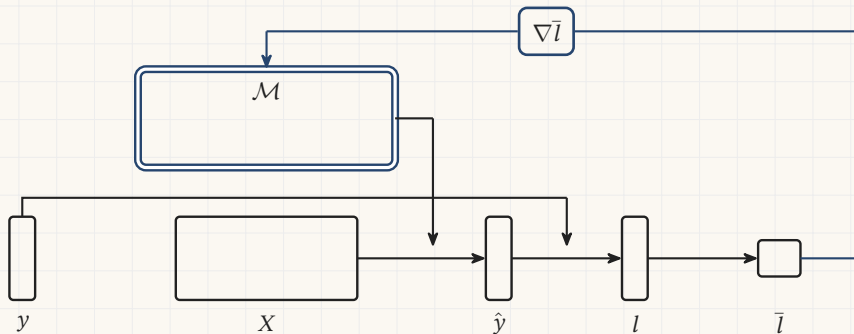
The five flats and the network are the ones from weekend 1.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

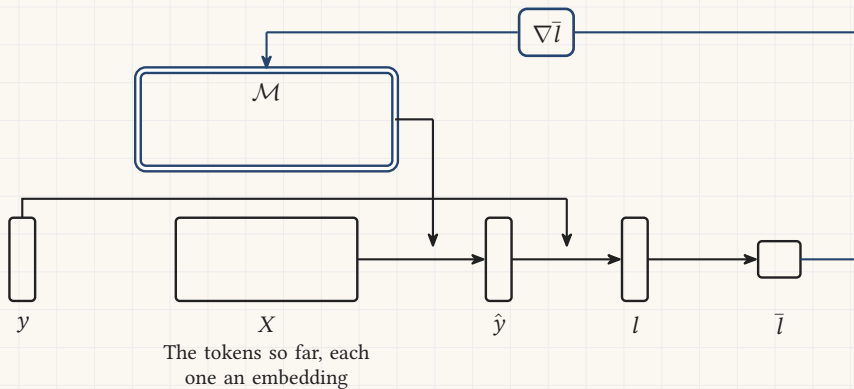
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

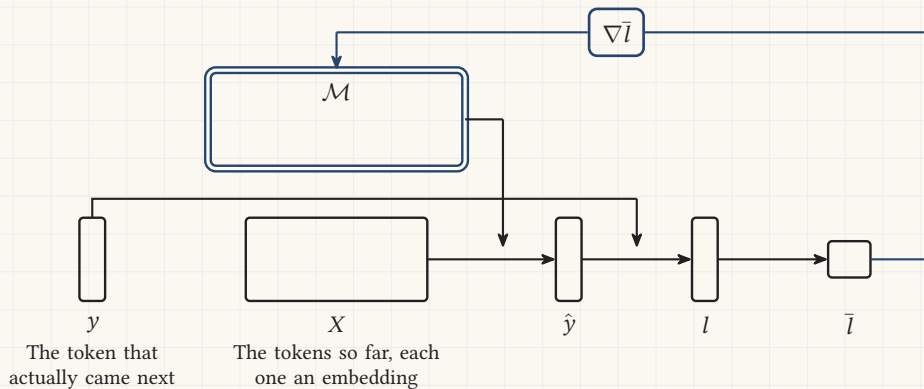
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

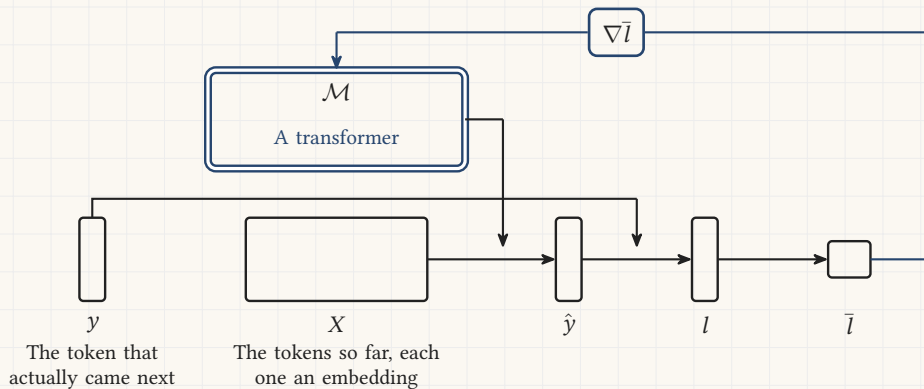
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

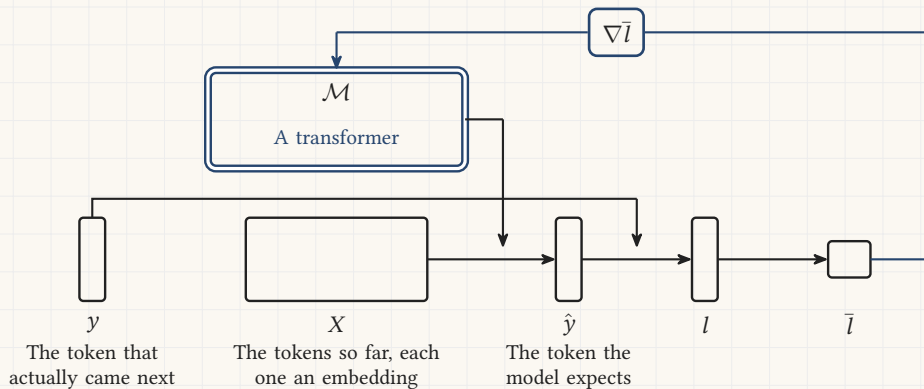
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



Large language models

A row is text with its next token cut off. Every position of every document gives one.

The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



2

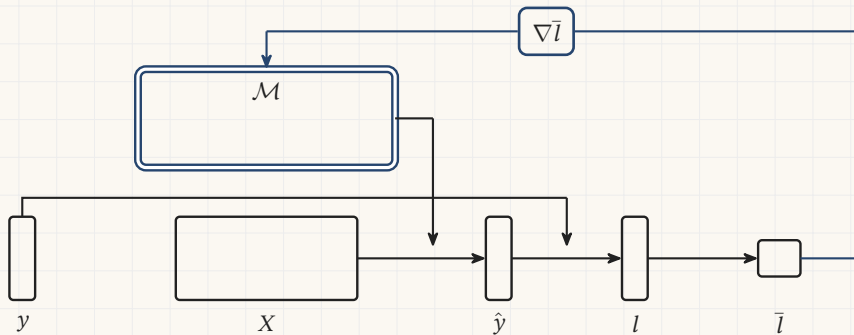
Section 2

How a language model becomes a chatbot

Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

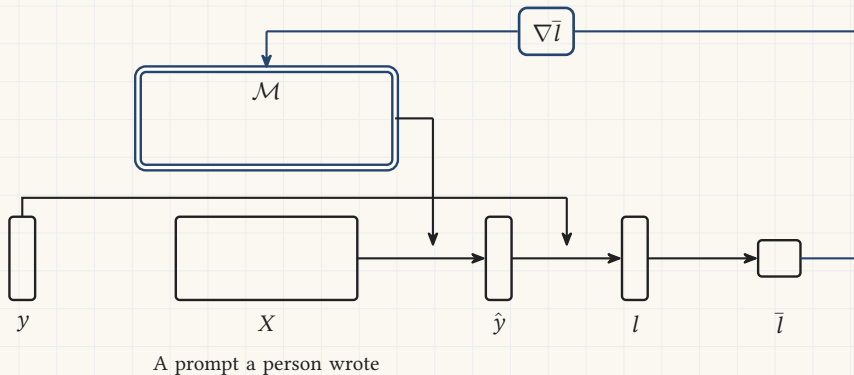
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

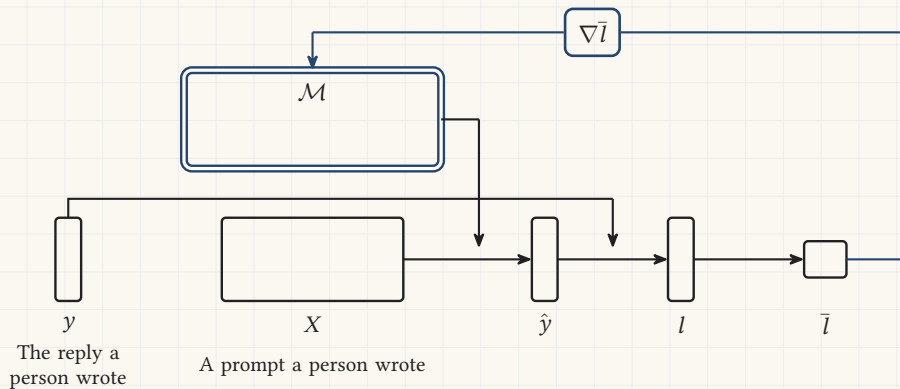
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

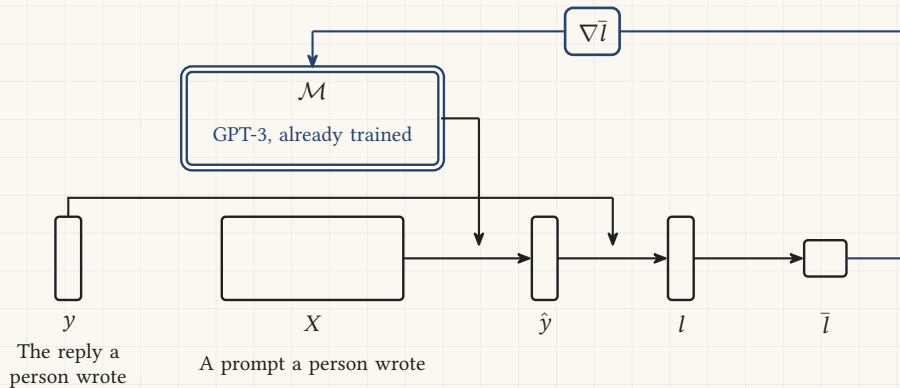
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

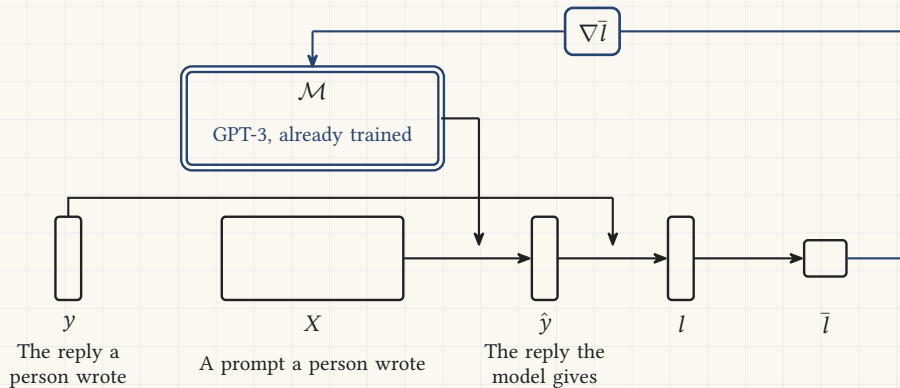
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

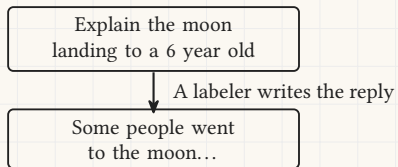
Explain the moon
landing to a 6 year old

The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration



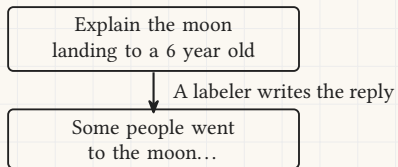
The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

12 725 prompts, most of them written by the labelers



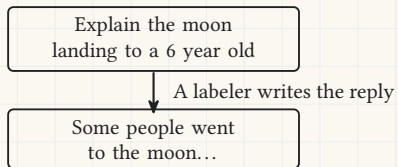
The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

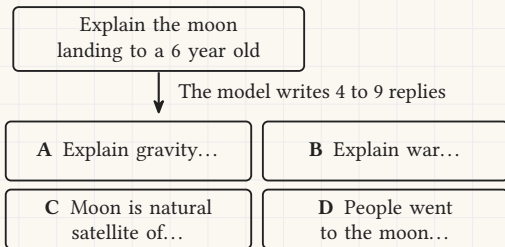
Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

12 725 prompts, most of them written by the labelers



One comparison



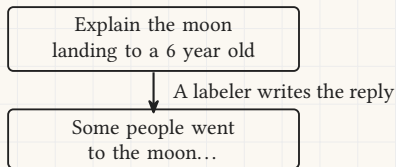
The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

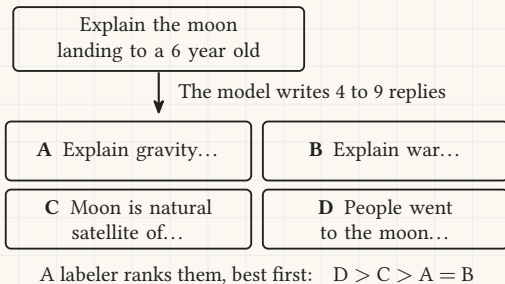
Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

12 725 prompts, most of them written by the labelers



One comparison



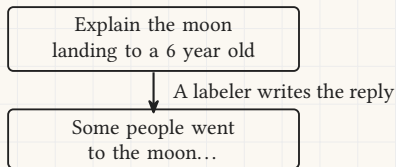
The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

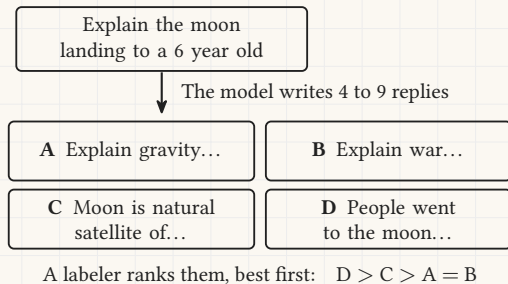
One demonstration

12 725 prompts, most of them written by the labelers



One comparison

33 207 prompts, most of them sent by users of the API



3

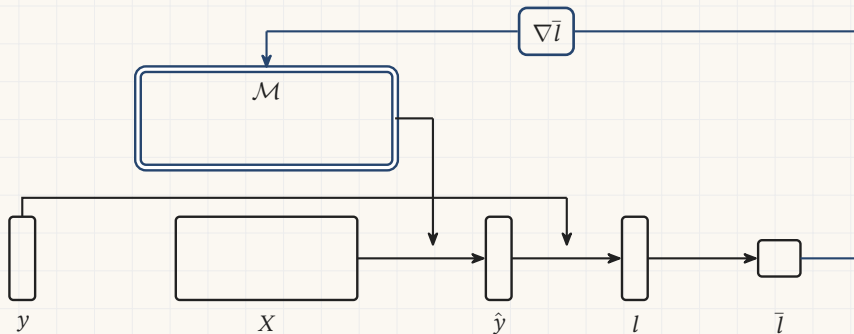
Section 3

How a chatbot becomes a reasoning model

Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

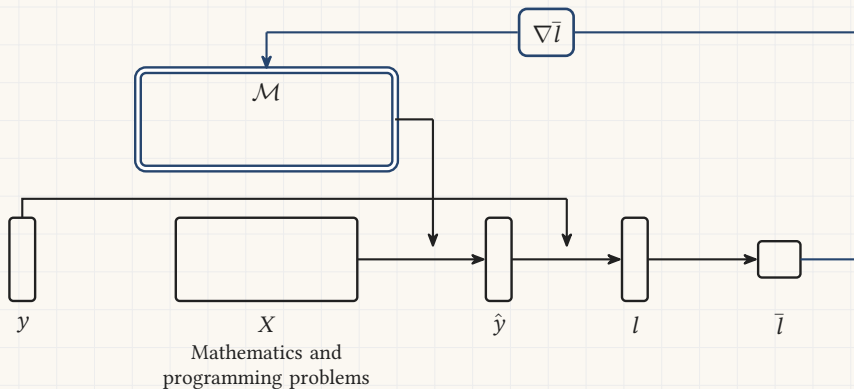
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

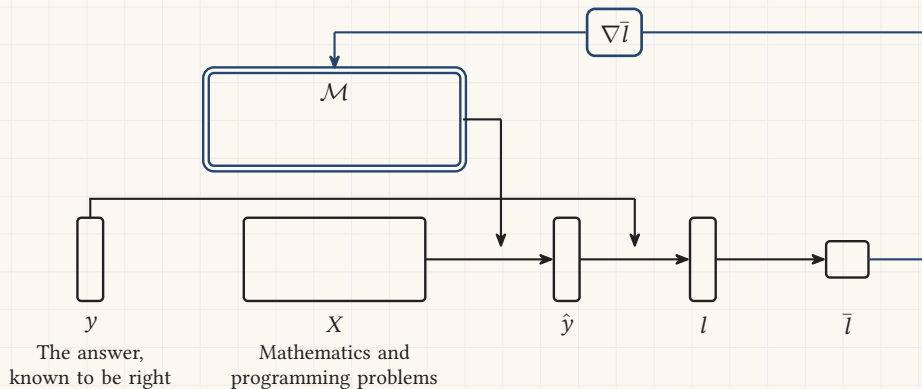
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

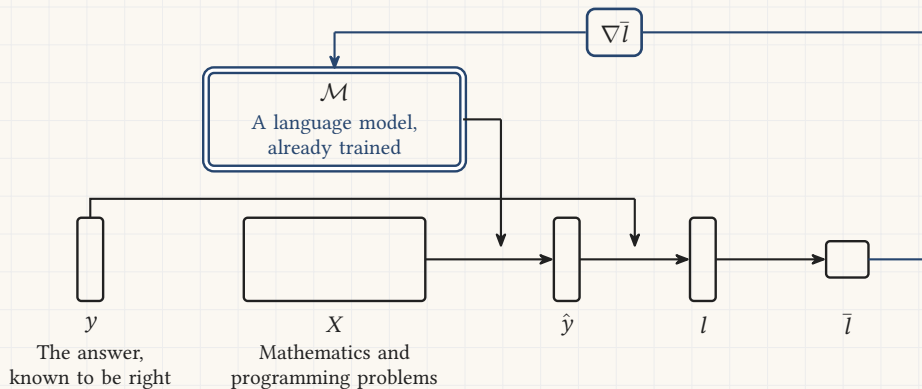
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

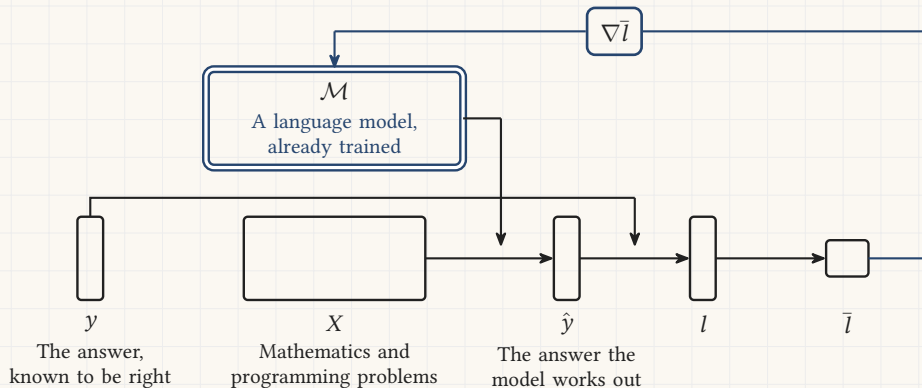
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



4

Section 4

Reinforcement learning

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

What is 12 times 7?

12

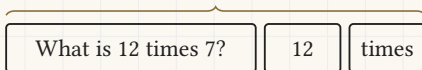
times

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

The state: the prompt and the tokens written so far

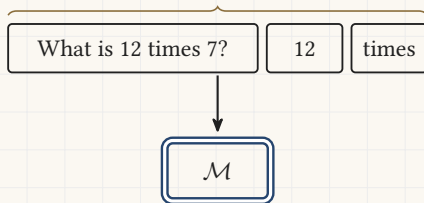


A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

The state: the prompt and the tokens written so far

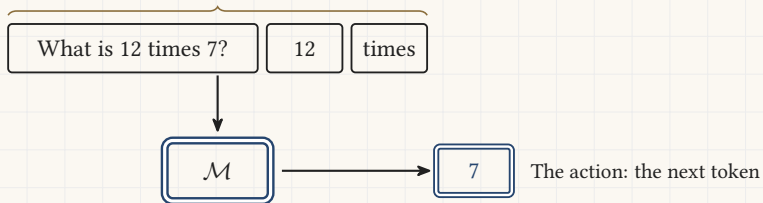


A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

The state: the prompt and the tokens written so far

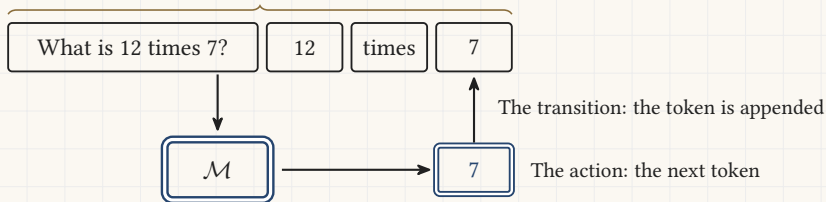


A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

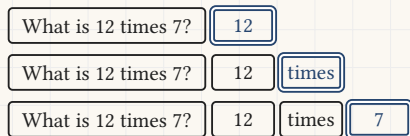
The state: the prompt and the tokens written so far



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

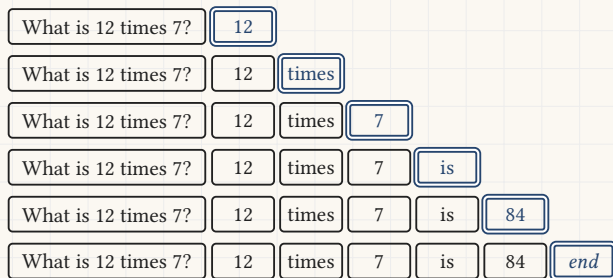
DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

What is 12 times 7?	12				
What is 12 times 7?	12	times			
What is 12 times 7?	12	times	7		
What is 12 times 7?	12	times	7	is	
What is 12 times 7?	12	times	7	is	84

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	end

0 while the answer is still being written

The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward	
What is 12 times 7?	12					0	
What is 12 times 7?	12	times				0	
What is 12 times 7?	12	times	7			0	0 while the answer is still being written
What is 12 times 7?	12	times	7	is		0	
What is 12 times 7?	12	times	7	is	84	0	
What is 12 times 7?	12	times	7	is	84	1	1 at the end: 84 is right. A wrong answer earns 0
The end-of-answer token: the episode ends							

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	1
The end-of-answer token: the episode ends						1
						1 The sum of the rewards

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$w \leftarrow w$
The weights

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow w - \underbrace{\alpha}_{\text{Minus a small step}}$$

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow w \underbrace{- \alpha}_{\text{Minus a small step}} \underbrace{\nabla}_{\text{The direction that increases}}$$

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Minus a small step}} - \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{L(w)}_{\text{The loss}}$$

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$w \leftarrow w$
The weights

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$$\underbrace{w}_{\text{The weights}} \leftarrow w + \underbrace{\alpha}_{\text{Plus a small step}}$$

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w + \alpha}_{\text{Plus a small step}} \underbrace{\nabla}_{\text{The direction that increases}}$$

Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

$$\underbrace{w}_{\text{The weights}} \leftarrow \underbrace{w}_{\text{Plus a small step}} + \underbrace{\alpha}_{\text{The direction that increases}} \underbrace{\mathbb{E}[r_1 + \dots + r_T]}_{\text{The expected sum of rewards}}$$

References

Ashish Vaswani and others. Attention is all you need. *Advances in Neural Information Processing Systems* 30, 2017.
arxiv.org/abs/1706.03762

Long Ouyang and others. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems* 35, 27730 to 27744, 2022.
arxiv.org/abs/2203.02155

DeepSeek-AI. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645, 633 to 638, 2025.
doi.org/10.1038/s41586-025-09422-z

Zichen Liu and others. Understanding R1-Zero-like training: a critical perspective. *Conference on Language Modeling*, 2025.
arxiv.org/abs/2503.20783

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press, 2018. Chapter 13.
incompleteideas.net/book/RLbook2020.pdf